Santa's Workshop Tour Optimization

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1. Problem Definition

Goal:

- Assign each family to a day minimizing total cost

Objective Function:

$$f(S) = \text{Preference Cost} + \text{Accounting Penalty}$$

Hard Constraints:

- Daily occupancy between limits (total number of people attending the workshop each day must be between 125 and 300)

Soft Constraints:

- Family preferences (ranked choices)

2. Related Work

Sources Explored:

- Kaggle Leaderboard Solutions
- YouTube: Simulated Annealing

References:

- Kaggle Discussion Thread
- Streamlining Scheduling with Simulated Annealing

Common Techniques:

- Mixed Integer Linear Programming (most common and winner of the competition)
- Simulated Annealing

3. Solution Representation

Solution (State):

$$S = (d_1, d_2, ..., d_n)$$

- d_i = assigned day for family i

Data Structures (Implementation):

- **placement_dict:** family $\rightarrow$ assigned day
- **occupancy_dict:** day $\rightarrow$ number of people
- **choice_dict:** family $\rightarrow$ preferred days list

Initialization:

- Greedy assignment based on preferences

4. Neighborhood & Operators

Neighborhood Function:

- Select a family i
- Assign a different day from its preference list

$$S' = (d_1, ..., d_i', ..., d_n)$$

Mutation (Simulated Annealing):

- Random reassignment of one family

Crossover (Not Used):

- Not applicable (single-solution metaheuristic)

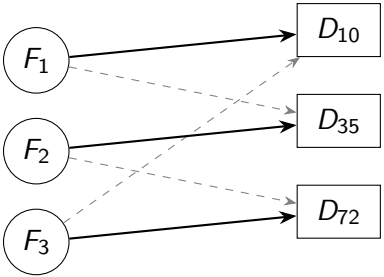
Transition Acceptance:

$$P = e^{-\Delta/T}$$

5. State as a Bipartite Graph

State Representation:

- A state is a valid set of edges in a bipartite graph
- Left partition: family nodes $(F_1, \ldots, F_n)$
- Right partition: day-choice nodes (possible assigned days)
- Each family has exactly one selected edge to one day node



Interpretation:

- Solid edges = chosen assignments defining the state
- Dashed edges = alternative candidate assignments
- Validity requires occupancy constraints on each day

6. Evaluation Function

Total Cost:

$$f(S) = C_{pref} + C_{account}$$

Preference Cost:

- Santa provides consolation gifts according to each family's assigned day relative to their preference rank
- The sum of these gifts over all families is the preference cost (C_{pref})
- choice_0: no consolation gifts
- choice_1: one $50 gift card to Santa's Gift Shop
- choice_2: one $50 gift card, and 25% off Santa's Buffet (value $9) for each family member
- choice_3: one $100 gift card, and 25% off Santa's Buffet (value $9) for each family member
- choice_4: one $200 gift card, and 25% off Santa's Buffet (value $9) for each family member

6. Evaluation Function (cont.)

- choice_5: one $200 gift card, and 50% off Santa's Buffet (value $18) for each family member
- choice_6: one $300 gift card, and 50% off Santa's Buffet (value $18) for each family member
- choice_7: one $300 gift card, and free Santa's Buffet (value $36) for each family member
- choice_8: one $400 gift card, and free Santa's Buffet (value $36) for each family member
- choice_9: one $500 gift card, and free Santa's Buffet (value $36) for each family member, and 50% off North Pole Helicopter Ride tickets (value $199) for each family member
- otherwise: one $500 gift card, and free Santa's Buffet (value $36) for each family member, and free North Pole Helicopter Ride tickets (value $398) for each family member

6. Evaluation Function (cont.)

Accounting Penalty:

- Extra empirical cost based on occupancy and day-to-day variation:

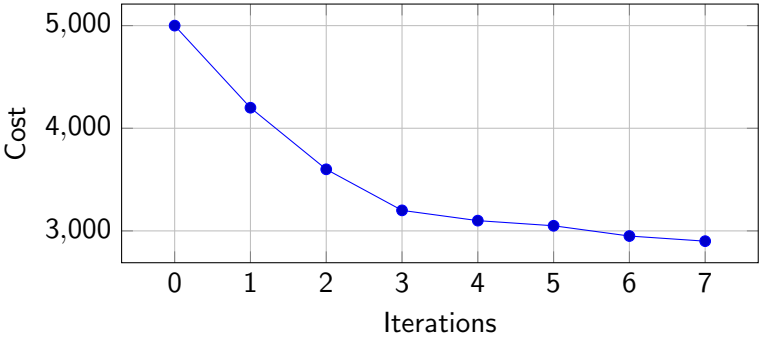
$$\sum_{d=100}^{1} \frac{N_d - 125}{400} N_d^{\left(0.5 + \frac{|N_d - N_{d+1}|}{50}\right)}$$

- Where N_d is occupancy on day d, N_{d+1} is occupancy on the previous day, and for $d = 100$, $N_{101} = N_{100}$

Hard Constraint Handling:

- Invalid states are rejected
  - The total number of people attending the workshop each day must be between 125 and 300

7. Simulated Annealing Behavior



Behavior:

- Early exploration (high temperature)
- Later exploitation (low temperature)

8. Implementation

Technology:

- Python + Pandas

Approach:

- Greedy initialization
- Iterative Simulated Annealing loops

Key Features:

- Efficient cost recomputation
- Dictionary-based state tracking

Next Steps:

- Tune cooling schedule
- Improve neighborhood selection
- Try ILP formulation and MILP solvers